

| Criteria                                | Excellent (9–10 pts)                                                                                                                                                                      | Good (7–8 pts)                                                                                                   | Satisfactory (5–6 pts)                                                                                          | Needs Improvement (0–4 pts)                                                          |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|                                         | deep understanding of what each prompt requires.                                                                                                                                          | accurate but may lack depth or nuance.                                                                           |                                                                                                                 | irrelevant, or show misunderstanding.                                                |
| <b>Task 2: Create Research Question</b> | Research question is sophisticated, narrowly focused, argument-driven, and perfectly formatted. It clearly stems from the funnel process and is appropriate for university-level inquiry. | Research question is clear, arguable, and follows the funnel. It may be slightly broad or could be more precise. | Research question is too broad, factual, or not properly argument-driven. The funnel process is weakly applied. | Research question is missing, purely descriptive, or does not follow the assignment. |
| <b>Task 3: “So What?” Test</b>          | The “So What?” explanation is insightful, compelling, and clearly articulates the academic or real-world significance of answering the question.                                          | The “So What?” is logical and relevant, explaining why the question matters.                                     | The “So What?” is vague, superficial, or only loosely connected to the question.                                | The “So What?” is missing, illogical, or fails to justify the question's importance. |

## Example of a Completed Homework Assignment

### Task 1: Deconstruct Prompts

- Prompt:** "analyse how the narrative structure of a non-linear novel contributes to its central themes."
  - Command Word:** analyse
  - Paraphrase:** I need to break down the complex, non-chronological structure of a novel and explain in detail how this specific way of telling the story shapes, emphasises, and deepens the book's main ideas.
- Prompt:** "Evaluate the effectiveness of international sanctions as a tool for compelling political change."
  - Command Word:** Evaluate
  - Paraphrase:** I need to make a judgment about how well sanctions work, using specific criteria (e.g., economic impact, political outcome) and support my judgment with historical or contemporary examples.

### Task 2: Create Research Question

**My Chosen Topic:** Education Technology

**My Research Question:** To what extent do AI-powered personalised learning platforms actually improve long-term knowledge retention in secondary school mathematics, compared to traditional teacher-led instruction?

### Task 3: “So What?” Test

**Why it matters:**

Answering this question is crucial because schools and governments are investing billions in EdTech without robust evidence of its long-term educational value. If AI platforms are proven to be more effective, it could revolutionise resource allocation and teacher training. If not, it would challenge the current trend of digitalization in classrooms and refocus investment on teacher support and traditional pedagogies. Ultimately, this research matters because it directly impacts how future generations learn fundamental skills.

## Chapter 4: The Power of Good Logic

第四章：逻辑的力量 I (Luójí de lìliàng I)

**Your Goal for This Chapter 本章目标 (Běn zhāng mùbiāo):** To identify the most common logical fallacies that weaken arguments. You will learn to spot flawed reasoning in others' writing and avoid it in your own.

识别削弱论证的最常见逻辑谬误。你将学会在别人的写作中发现有缺陷的推理，并在自己的写作中避免它。

## Lesson 4.1: Why Logic Matters

第一课：为什么逻辑重要 (Wèishéme luójí zhòngyào)

### What is logic?

At its core, **logic is the study of correct reasoning**. It provides a systematic framework for distinguishing good arguments from bad ones.

Think of it as the "science of correct thinking."

It doesn't primarily focus on what people *actually* think (that's psychology) or what is *factually true* (that's the sciences), but rather on whether a conclusion *comes from a correct line of thinking*.

### Part 1: The Architecture of Argument

论证的架构 (Lùnzhèng de jiàgòu)

Think back to our metaphor: writing is architecture. If your **evidence 证据 (zhèngjù)** and **explanation 解释 (jiěshì)** are the bricks and mortar of your paragraph, then **logic 逻辑 (luójí)** is the foundation. It's what holds your entire argument together and makes it strong enough to withstand criticism.

**Weak logic is like a weak foundation. Your whole argument can collapse.**

### Part 2: What is a Logical Fallacy?

什么是逻辑谬误？(Shénme shì luójí miùwù?)

A **logical fallacy 逻辑谬误 (luójí miùwù)** is an error in thinking. It's a flaw in the structure of an argument that makes it invalid, misleading, or unfair. Fallacies often seem persuasive because they appeal to our emotions or biases, but they don't provide good evidence for a claim.

Learning about fallacies is like learning the tricks of a magician. Once you know how the trick works, you can't be fooled by it anymore.

## Common types of faulty logic (logical fallacies).

### 1. Hasty Generalisation

**Definition:** Drawing a broad conclusion based on a small, unrepresentative, or insufficient sample size.

- **Bad Logic:** "My two roommates from Country X were messy, so everyone from Country X is messy."
- **Why it's Faulty:** It makes a sweeping generalization about an entire population based on only two examples, which are not necessarily representative.

### 2. Ad Hominem (Against the Person)

**Definition:** Attacking the character, motive, or other personal attributes of the person making an argument instead of addressing the substance of the argument itself.

- **Bad Logic:** "Dr. Smith argues for climate change action, but he was divorced last year, so his opinion can't be trusted."
- **Why it's Faulty:** The personal life of the speaker is irrelevant to the scientific evidence for climate change. The argument is dismissed based on who said it, not on its merits.

### 3. Straw Man

**Definition:** Misrepresenting, exaggerating, or oversimplifying an opponent's argument to make it easier to attack.

- **Bad Logic:**
  - Person A: "We should consider more funding for renewable energy sources."
  - Person B: "So you want to shut down all coal plants tomorrow and leave thousands unemployed? That's a terrible plan."
- **Why it's Faulty:** Person B has distorted Person A's moderate proposal into an extreme, unrealistic position that is easy to criticize.

#### 4. False Cause (Post Hoc Ergo Propter Hoc)

**Definition:** Assuming that because one event occurred after another, the first event must have caused the second.

- **Bad Logic:** "I wore my lucky socks and then aced my exam. Therefore, my lucky socks caused me to ace the exam."
- **Why it's Faulty:** It confuses correlation with causation. The two events happened sequentially, but there is no proven causal link; studying was the more likely cause.

#### 5. Slippery Slope

**Definition:** Arguing that a relatively small first step will inevitably lead to a chain of related (and extreme) negative events.

- **Bad Logic:** "If we allow students to redo this assignment, next they'll demand to redo every assignment, and then they'll expect to pass the course without doing any work, completely destroying academic standards."
- **Why it's Faulty:** It presents a series of hypothetical events as an inevitable outcome without evidence for each step in the chain.

#### 6. Appeal to Authority

**Definition:** Using the opinion of an authority figure as evidence for an argument in an area outside of their expertise.

- **Bad Logic:** "A famous actor and action star endorses this brand of vitamins, so it must be the most scientifically effective."
- **Why it's Faulty:** The actor is an authority on acting, not on nutrition or biochemistry. Their endorsement is not valid evidence for the product's efficacy.

#### 7. False Dilemma (Either/Or Fallacy)

**Definition:** Presenting only two possible options or outcomes when in reality there are more nuances or alternatives.

- **Bad Logic:** "You're either with us in this war, or you're with the terrorists."
- **Why it's Faulty:** It frames a complex geopolitical issue as a binary choice, ignoring the possibility of neutral stances, diplomatic solutions, or other alternatives.

#### 8. Circular Reasoning (Begging the Question)

**Definition:** An argument where the conclusion is implicitly or explicitly assumed in one of the premises. It restates the claim rather than proving it.

- **Bad Logic:** "The Bible is the word of God because it says so in the Bible."
- **Why it's Faulty:** The argument offers no independent evidence; it simply uses the claim itself as proof for the claim.

#### 9. Red Herring

**Definition:** Introducing an irrelevant topic or piece of evidence to divert attention from the original issue.

- **Bad Logic:**
  - Person A: "The mayor's economic policy has failed to reduce unemployment."
  - Person B: "But let's not forget that the mayor volunteers at an animal shelter on weekends. He's a good man."

- **Why it's Faulty:** The mayor's character is a distraction from the specific criticism of his economic policy.

## 10. Appeal to Emotion

**Definition:** Manipulating an emotional response (pity, fear, anger, nationalism) in place of a valid or compelling argument.

- **Bad Logic:** "This university should not cut the arts program. Think of the sad, disappointed children who will never get to paint again."
- **Why it's Faulty:** While the emotional appeal is powerful, a strong argument should focus on the educational value, cost-benefit analysis, or statistical impact of the arts program, not just the emotional reaction to its loss.

## 11. Bandwagon Fallacy (Appeal to Popularity)

**Definition:** Arguing that a claim must be true or good simply because it is popular or many people believe it.

- **Bad Logic:** "Millions of people believe that crystal healing works, so it must be a valid medical treatment."
- **Why it's Faulty:** Popularity is not proof. The truth of a claim is determined by evidence and logic, not by the number of people who believe it.

By recognising and avoiding these common fallacies, you can strengthen your academic writing, making your arguments more logical, credible, and persuasive.

## Lesson 4.2: Spotting Two Common Fallacies

第二课：识别两种常见谬误 (Shíbié liǎng zhǒng chángjiàn miùwù)

### Part 1: Fallacy #1 - Hasty Generalization

谬误 #1 - 草率结论 (Cǎoshuài jiélùn)

**What it is 是什么:** Making a broad claim based on a very small or unrepresentative sample. You are jumping to a conclusion without enough evidence.

基于一个非常小或不具代表性的样本得出一个广泛的主张。你在没有足够证据的情况下草率得出结论。

**In simple terms 简单来说:** Stereotyping. Assuming what is true for one is true for all.

刻板印象。认为对一个人是真的，对所有人也是真的。

#### Everyday Example 日常例子:

- "I met two rude people from City X. Everyone from City X must be rude."  
“我遇到了两个来自X市的粗鲁的人。所有来自X市的人一定都很粗鲁。”
- "This one study says this product works. Therefore, it is proven to work for everyone."  
“这一项研究说这个产品有效。因此，它被证明对所有人都有效。”**How to avoid it 如何避免:** Use qualifiers like "may," "often," "sometimes," or "in some cases." Make sure your evidence is sufficient and representative.  
使用限定词，如“可能”、“通常”、“有时”或“在某些情况下”。确保你的证据是充分且具有代表性的。

## Resources

### Category 1: Educational Resources (Defining the Fallacy with Clear Examples)

These are excellent for understanding the core concept with simple, constructed examples.

1. [YourLogicalFallacyIs.com](http://YourLogicalFallacyIs.com): Hasty Generalization

- **Link:** <https://yourlogicalfallacyis.com/hasty-generalization>
- **What it demonstrates:** This site provides a very concise, easy-to-understand definition and a simple, classic example: "Someone who meets three people from a foreign country and finds them to be quiet and reserved may then conclude that all people from that country are quiet and reserved."

## 2. The Fallacy Files: Hasty Generalization

- **Link:** <http://www.fallacyfiles.org/hastygen.html>
- **What it demonstrates:** This resource goes into more philosophical depth, explaining the difference between inductive generalization and the fallacious "hasty" version. It uses historical and literary examples.

## 3. Purdue Online Writing Lab (OWL): Logical Fallacies

- **Link:** [https://owl.purdue.edu/owl/general\\_writing/academic\\_writing/logic\\_in\\_argumentative\\_writing/fallacies.html](https://owl.purdue.edu/owl/general_writing/academic_writing/logic_in_argumentative_writing/fallacies.html)
- **What it demonstrates:** While it covers many fallacies, it includes Hasty Generalization (sometimes under "Faulty Generalization") with clear academic examples, such as making a broad claim based on a small or unrepresentative sample.

## Category 2: Analysis and Opinion Pieces (Real-World Examples)

These articles analyze real-world rhetoric and point out where Hasty Generalization occurs, often in politics, media, and social discourse.

### 1. The Guardian: "How to spot a hasty generalization – and how to stop making them"

- **Link:** <https://www.theguardian.com/lifeandstyle/2023/mar/26/how-to-spot-a-hasty-generalisation-and-how-to-stop-making-them>
- **What it demonstrates:** This article directly addresses the fallacy in everyday life, discussing how we use it about places, groups of people, and even in our self-talk. It's a practical guide to recognizing the pattern.

### 2. Psychology Today: "The Hasty Generalization Fallacy"

- **Link:** <https://www.psychologytoday.com/us/blog/thoughts-thinking/202009/the-hasty-generalization-fallacy>
- **What it demonstrates:** This piece connects the logical fallacy to cognitive psychology, explaining how our brains are wired to make quick generalizations and how this can lead to errors in judgment, stereotypes, and prejudice.

### 3. Logically Fallacious: Hasty Generalization Entry

- **Link:** <https://www.logicallyfallacious.com/logicalfallacies/Hasty-Generalization>
- **What it demonstrates:** This site is a comprehensive encyclopedia of fallacies. Its entry on Hasty Generalization provides multiple real-world examples, such as "My grandfather smoked four packs of cigarettes a day since age fourteen and lived until age ninety-two. Therefore, smoking really can't be that bad for you."

## Category 3: Video Explanations (For Visual Learners)

Sometimes, seeing the fallacy in action is the best way to learn.

### 1. Khan Academy: "Hasty Generalization"

- **Link:** <https://www.khanacademy.org/partner-content/wi-phi/critical-thinking/v/hasty-generalization>
- **What it demonstrates:** This short, animated video clearly explains the fallacy with simple examples, making it perfect for students or anyone new to the concept.

## How to Find More Examples Yourself:

- **Search News Sites:** Look on sites like **The New York Times**, **The Atlantic**, or **BBC** for articles analyzing political speeches, marketing claims, or social media trends. Opinion and analysis sections are where you'll most often find fallacy breakdowns.
- **Use Specific Search Terms:** Try searches like:
  - "Hasty generalization in politics example"

- "News article about stereotyping as hasty generalization"
- "Analysis of [Political Figure's] speech logical fallacies"
- **Fact-Checking Websites:** Sites like [Snopes.com](https://snopes.com) or [PolitiFact.com](https://politiFact.com) often deconstruct claims by pointing out flawed reasoning, including hasty generalizations based on insufficient evidence.

By reviewing these resources, you'll see that Hasty Generalization is everywhere, from a casual conversation about "all politicians" to a news segment making a sweeping claim about a generation. The key is recognizing when a conclusion is drawn from a sample that is too small, biased, or unrepresentative to support it.

## Part 2: Fallacy #2 - Loaded Language (LL)

谬误 #2 - 煽动性语言 (Shāndòng xìng yǔyán)

**What it is 是什么:** Using words with strong positive or negative emotions (connotations) to manipulate an audience instead of using facts and reason.

使用带有强烈积极或消极情绪（内涵）的词语来操纵听众，而不是使用事实和理性。

**In simple terms 简单来说:** Using slanted or biased words to make something seem better or worse than it is.

使用带有倾向性或偏见的词语，使某事看起来比实际更好或更糟。

**Everyday Example 日常例子:**

- **Logic Fallacy:** "My opponent supports the *disastrous* and *heartless* policy that will *destroy our community*."  
**Correct Logic:** (Instead of: "My opponent supports the policy, which I believe will have negative consequences.")  
“我的对手支持将会摧毁我们社区的灾难性且无情的政策。”（而不是：“我的对手支持该政策，我认为这会带来负面后果。”）
- "We must pass this *common-sense*, *patriotic* bill for the *sake of our children*."  
(Instead of presenting the bill's actual benefits.)  
“为了我们的孩子，我们必须通过这项常识性的、爱国的法案。”（而不是展示法案的实际好处。）

**How to avoid it 如何避免:** Use neutral, factual language. Let your **evidence** create feeling, not your **adjectives**.

使用中立的、事实性的语言。让你的证据来创造感觉，而不是你的形容词。

## Part 3: Practice: Spot the Fallacy

练习：发现谬误 (Liànxí: Fāxiàn miùwù)

**Instructions 说明:** For each statement below, identify which fallacy being used: **Hasty Generalisation (HG)** or **Loaded Language (LL)**.

对于下面的每个陈述，识别所使用的谬误：**草率结论 (HG)** 或 **煽动性语言 (LL)**。

- "I got food poisoning from one sushi restaurant. I'm never eating sushi again."  
“我在一家寿司店食物中毒了。我再也不吃寿司了。”  
• **Fallacy 谬误:** \_\_\_\_\_
- "The politician's *radical*, *un-American* proposal will *crush the dreams* of hardworking families."  
“这位政客激进的、非美国的提案将粉碎勤劳家庭的梦想。”  
• **Fallacy 谬误:** \_\_\_\_\_
- "Everyone knows that social media is a toxic waste of time."  
“每个人都知道社交媒体是一种有毒的浪费时间。”  
• **Fallacy 谬误:** \_\_\_\_\_ (Hint 提示: Which words are "loaded"? 哪些词是“煽动性的”?)
- "My grandfather smoked a pack a day and lived to be 100. Therefore, smoking isn't bad for you."  
“我的祖父每天抽一包烟，活到了100岁。因此，吸烟对你没有坏处。”

- Fallacy 谬误: \_\_\_\_\_

## Lesson 4.3: Applying Logic to Your Writing

第三课：将逻辑应用于写作 (Jiāng luójí yīngyòng yú xiězuò)

### Part 1: Revise the Fallacies

修改谬误 (Xiūgǎi miùwù)

Now, let's fix the examples from the previous exercise. Rewrite each statement to make it logical and fair.

现在，让我们修改上一练习中的例子。重写每个陈述，使其变得逻辑且公平。

1. **Original 原句:** "I got food poisoning from one sushi restaurant. I'm never eating sushi again."

**Revised 修改后:** \_\_\_\_\_

2. **Original 原句:** "The politician's *radical, un-American* proposal will *crush the dreams* of hardworking families."

**Revised 修改后:** \_\_\_\_\_

### Part 2: Analyze a Real-World Argument

分析现实世界的论证 (Fēnxī xiànrshí shìjiè de lùnzhèng)

Find a short opinion piece, social media post, or comment online.

找一篇短评、社交媒体帖子或在线评论。

**Paste or describe it here 在这里粘贴或描述它:**

**Analyze it 分析它:**

- Does it contain **Hasty Generalisation**? How? 它是否包含**草率结论**? 如何体现的?
- Does it use **Loaded Language**? Which words? 它是否使用了**煽动性语言**? 哪些词?
- Is the argument logically strong? Why or why not? 这个论证在逻辑上强大吗? 为什么?

## Homework: Fallacy Detective

作业：谬误侦探 (Miùwù zhēntàn)

### Task 1: Find and Identify 发现与识别

Over the next day, be a "Fallacy Detective." Listen to conversations, read headlines, or scroll through social media.

在接下来的一天里，成为一名“谬误侦探”。倾听对话，阅读头条新闻，或浏览社交媒体。

Find **one example** of **Hasty Generalisation** and **one example** of **Loaded Language**. Write them down below.

找到一个**草率结论**的例子和一个**煽动性语言**的例子。把它们写在下面。

- **Hasty Generalisation Example 草率结论例子:**
- **Loaded Language Example 煽动性语言例子:**

### Task 2: Revise and Improve 修改与改进

Choose one of the examples you found above and rewrite it to remove the logical fallacy and make it a stronger, more

factual statement.

选择你上面找到的一个例子，重写它，消除逻辑谬误，使其成为一个更强大、更符合事实的陈述。

- **My Revised Statement** 我修改后的陈述:

### Task 3: Reflect 反思

Why is it important to avoid these fallacies in your own academic writing?

在你自己学术写作中避免这些谬误为什么很重要？

## Example Homework: The Fallacy Detective – Grading Rubric & Example

This assignment is a **Coursework Assessment**. It will be graded on a scale of 0–10 points.

### Grading Rubric

| Criteria                            | Excellent (9–10 pts)                                                                                                                                                     | Good (7–8 pts)                                                                                     | Satisfactory (5–6 pts)                                                                       | Needs Improvement (0–4 pts)                                            |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Task 1: Find &amp; Identify</b>  | Both examples are clear, real-world instances of the specified fallacies. Identification is correct and includes a brief explanation of why the statement is fallacious. | Examples are relevant and correctly identified. Explanation is present but could be more detailed. | One example may be weak or incorrectly identified. Explanation is vague.                     | Examples are missing, incorrectly identified, or not actual fallacies. |
| <b>Task 2: Revise &amp; Improve</b> | Revised statements are logical, fair, and remove all fallacious reasoning. They maintain the original intent but use evidence-based, nuanced language.                   | Revised statements correct the fallacy and are logical. They may still be somewhat simplistic.     | Revision attempts to correct the fallacy but is incomplete or introduces new logical issues. | Revision is missing, or the statement remains fallacious.              |
| <b>Task 3: Reflect</b>              | Reflection demonstrates a deep, personal understanding of why avoiding fallacies is critical for academic credibility and ethical argumentation.                         | Reflection adequately explains the importance of avoiding fallacies in academic writing.           | Reflection is vague or generic, lacking personal connection or specific insight.             | Reflection is missing or irrelevant.                                   |

### Example of a Completed Homework Assignment

#### Task 1: Find and Identify

##### Hasty Generalization Example:

*Headline from social media:* "I saw two teenagers littering in the park today. Kids these days have no respect for the environment."

**Identification:** This is a Hasty Generalization. The speaker draws a sweeping conclusion about all "kids these days" based on the behaviour of only two individuals.

##### Loaded Language Example:

*From a political comment thread:* "My opponent's dangerous, radical plan will bankrupt hardworking families and destroy our community's way of life."